Serum Spike-in Projection Validation

GAIRA V5 — controlled perturbations through the frozen Raman atlas

Atlas	NMF k=24, FROZEN (fingerprint 09ed804a4083…, verified)
Perturbation spectra	4074 across 7 controlled datasets
Dose-response	ILS adenine (6 substrate×laser arms, 15 labs) + ergothioneine
Serum spikes	51 analytes × 5 replicates into pooled serum
Modality	Ag/Au-SERS — OUT OF DOMAIN for a Raman atlas by construction
Dose-response result	monotonic & saturating in 7/7 arms (ρ 0.93–1.00, p=0.002)
Serum-spike result	no overall directional agreement; 6 strong Ag adsorbers succeed
Atlas modified?	No — verified byte-identical before and after

PRINCIPAL FINDING

Concentration is registered; chemical identity mostly is not. Pure-analyte dose series move monotonically and saturate in every arm, far above a label-permutation null, and enzymatic depletion moves in the chemically correct direction. But serum spikes at physiological levels show no directional agreement with their own pure-analyte reference — except for strong silver adsorbers (hypoxanthine, xanthine, guanine, ergothioneine).

Executive summary

4074 spectra · atlas frozen and verified unchanged

Question

Does a known biochemical perturbation move a spectrum through the frozen Raman reference manifold in a physically meaningful, reproducible way? The atlas, its preprocessing, its basis and its ontology were frozen throughout; the study only projects.

A constraint that frames every result

Every controlled perturbation dataset in GAIRA is Ag- or Au-SERS, while the atlas is built from pure Raman references. All projections here are therefore OUT OF DOMAIN by construction (median OOD 0.15–0.28 versus in-domain Raman references). Per the study design this is not treated as failure; the question is whether motion remains internally coherent despite it.

What the data support

1. Dose-response is real and strong. In all seven arms (six ILS adenine substrate×laser combinations across 15 laboratories, plus ergothioneine) distance from the blank rises monotonically with concentration: Spearman $\rho = 0.93$–1.00 against a label-permutation null of ≈ 0.24, $p = 0.002$ in every arm.

2. The dose relationship is saturating, not linear. A Langmuir-type saturating model beat linear and logarithmic fits in 7/7 arms — the expected behaviour for analytes competing for a finite number of adsorption sites on a colloid surface.

3. Replicate displacements are reproducible: median direction cosine 0.71 across concentrations and 0.87 for the serum spikes.

4. Controls behave correctly. Unspiked serum replicates agree at 0.999; ILS blanks show no systematic batch drift (max 0.024); enzymatic urate depletion moves AWAY from the urate direction (cos −0.61), which is the chemically expected sign; and urate versus isotopically-labelled urate remain close (cos 0.87), as isotopologues should.

What the data do NOT support

5. Serum spikes do not, in general, move toward their own pure-analyte reference. Median cos(spike displacement, pure-SERS direction) = -0.012 versus a mismatched-analyte null of -0.036; only 7 of 53 analytes exceed the 95th percentile of that null. Chemical identity is largely NOT recovered from a physiological-level spike in serum.

6. Trajectories are monotonic in distance but not smooth in direction: mean consecutive-step cosine ranges from +0.43 to −0.26, i.e. several arms zig-zag while still advancing. Monotonic magnitude is not the same as a coherent path.

Interpretation (distinguished from the observations above)

The analytes that do succeed are chemically the ones that should: hypoxanthine (0.89), xanthine (0.74), guanine (0.41) — purines that chemisorb to silver through ring nitrogen lone pairs — and ergothioneine (0.53), whose thione sulfur binds silver strongly. Displacement magnitude and direction agreement correlate at $r = +0.54$. The consistent reading is that the atlas registers a perturbation when the analyte generates enough SERS signal to rise above the colloid background, and is blind to it otherwise. That is a property of the Ag-colloid measurement, not of the atlas.

Speculation (explicitly labelled)

If the limiting factor is surface competition rather than the manifold, then spikes measured at higher effective surface coverage — or with a background-suppressed acquisition — might recover directional agreement for weaker adsorbers. This study cannot test that claim and does not assert it.

Figure 1

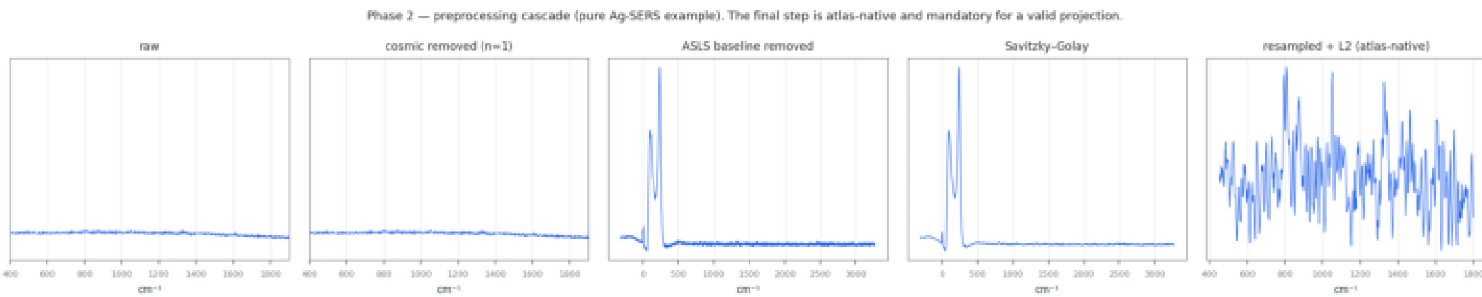

Figure 1 — Phase 2 preprocessing cascade. Cosmic-ray removal is applied only where the wavenumber axis oversamples the narrowest real band; on the 3 cm^{-1} ILS axis it is declined rather than risk deleting genuine signal.

Figure 2

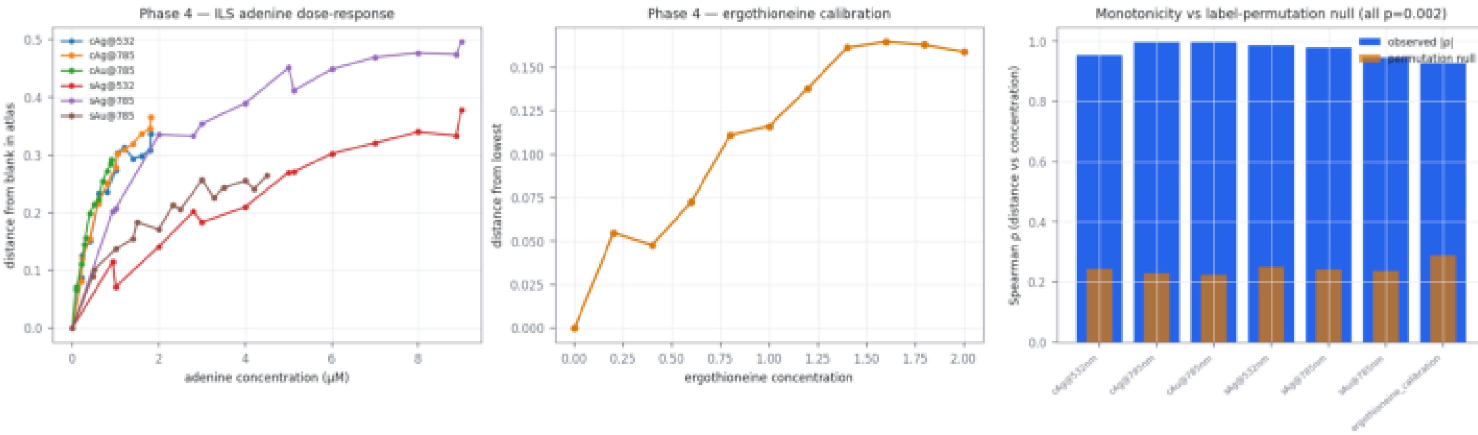

Figure 2 — Phase 4/8 dose-response. Distance from the blank rises monotonically and saturates in every arm, far above a label-permutation null.

Figure 3

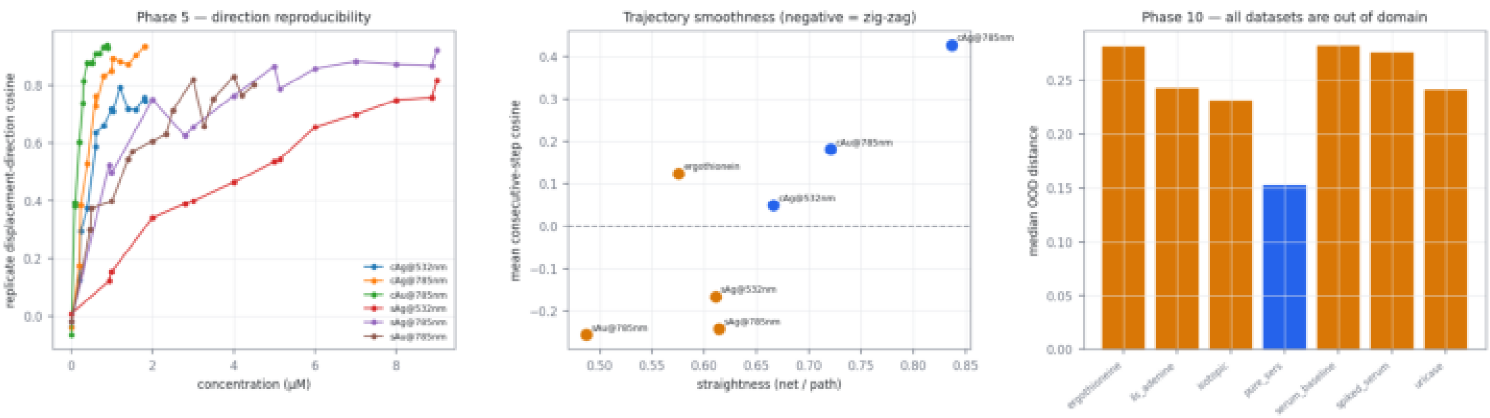

Figure 3 — Phase 5/10. Replicate direction reproducibility, trajectory smoothness (negative step cosine indicates zig-zag), and out-of-domain distance for every dataset.

Figure 4

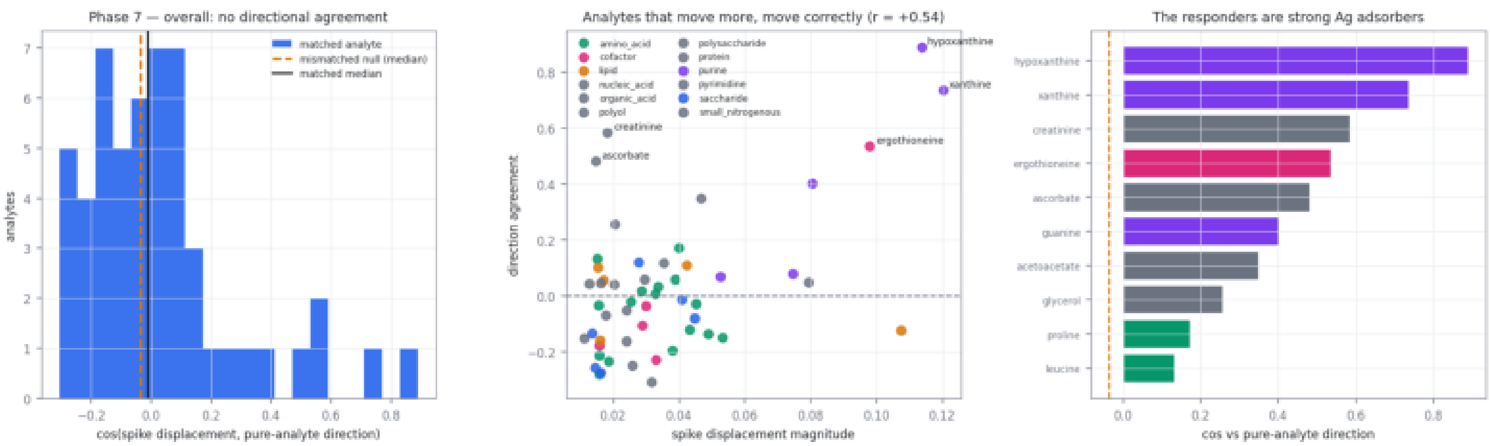

Figure 4 — Phase 7, the decisive test. Overall there is no directional agreement between a serum spike and its pure-analyte reference; the exceptions are strong silver adsorbers, and displacement magnitude predicts direction agreement.

Figure 5

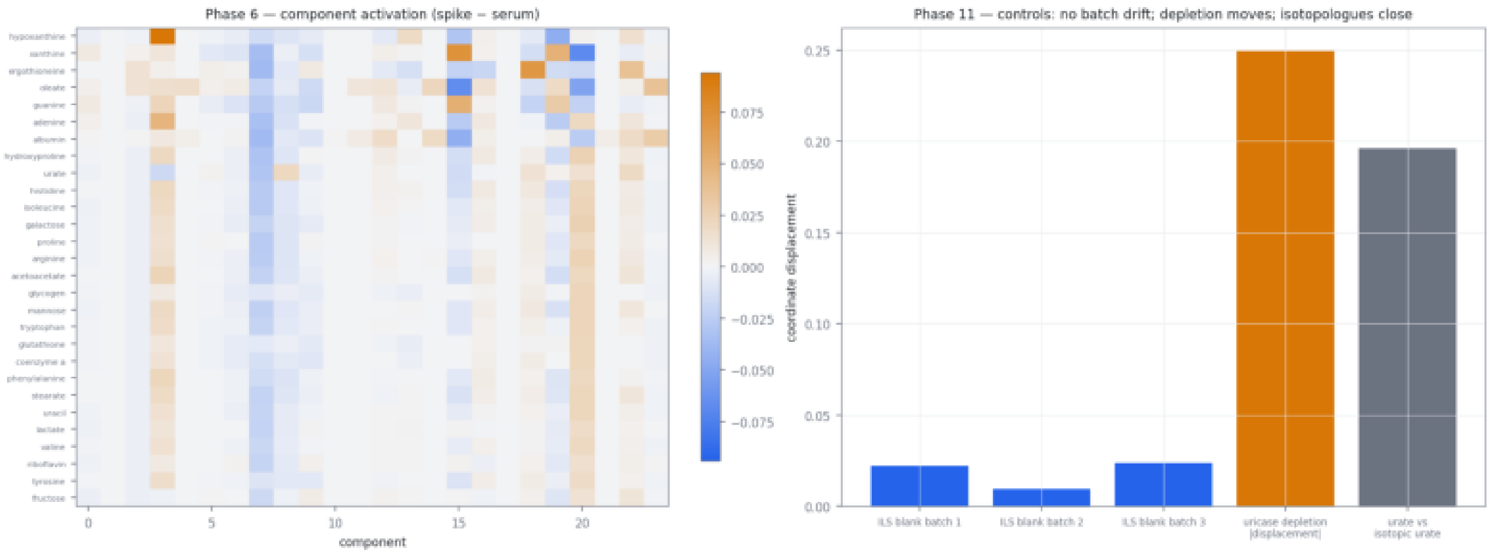

Figure 5 — Phase 6/11. Component activation per spiked analyte, and the control panel: no batch drift, depletion moves, isotopologues stay close.

Datasets and preprocessing

Phase 1 — dataset audit

dataset	n	analytes	levels	labs	substrates	lasers
ils_adenine	3381	1	43	15	cAg|cAu|sAg|sAu	532|785
spiked_serum	265	53	33	1	Ag colloid	785
serum_baseline	15	1	1	1	Ag colloid	785
pure_sers	265	53		1	Ag colloid	785
uricase	20	1	0	1	Ag colloid	785
isotopic	73	1	0	1	Ag colloid	785
ergothioneine	55	1	11	1	cAg	785

Phase 2 — preprocessing decisions

Projection into a frozen NMF basis is only meaningful in the representation the basis was fitted in, so the atlas-native pipeline (ASLS → Savitzky-Golay → L2 on 450–1800 cm^{-1} @ 2 cm^{-1}) is mandatory and was NOT tuned. Only the pre-steps were evaluated.

Cosmic-ray removal was initially over-aggressive: a naive large-residual rule flagged 13% of points on the 3 cm^{-1} ILS axis, i.e. it was deleting real bands. The rule was replaced by a sharpness test that is applied only when the axis oversamples the narrowest plausible band (≤2 cm^{-1}/point). On the ILS axis despiking is therefore declined and recorded as skipped; on the 1.7 cm^{-1} B&WTek axis it removes 0.09% of points.

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replicate cosine by dataset (median): ergothioneine=0.998, ils_adenine=0.831, isotopic=0.945, pure_sers=0.948, serum_baseline=0.998,
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455 replicate spectra were flagged as robust-z outliers versus their group median. They are documented in phase2_exclusions.csv and were retained in the analysis; no spectrum was removed on the basis of it being inconvenient.

Trajectories, mixtures and controls

Phase 4/8 — trajectory and dose-response model

experiment	lev	rho	p	null	straight	step-cos	model	linR2
ils_adenine::cAg@532nm	14	0.952	0.002	0.243	0.666	0.048	saturating	0.810
ils_adenine::cAg@785nm	14	0.996	0.002	0.230	0.837	0.426	saturating	0.890
ils_adenine::cAu@785nm	14	0.996	0.002	0.224	0.721	0.181	saturating	0.944
ils_adenine::sAg@532nm	14	0.987	0.002	0.250	0.611	-0.167	saturating	0.932
ils_adenine::sAg@785nm	14	0.978	0.002	0.241	0.614	-0.243	saturating	0.773
ils_adenine::sAu@785nm	15	0.946	0.002	0.235	0.487	-0.256	saturating	0.853
ergothioneine_calibration	11	0.927	0.002	0.289	0.575	0.123	saturating	0.902

Phase 9 — mixture behaviour

Each analyte was spiked into serum individually at a single concentration; no combinatorial A+B spike exists in this corpus, so Delta(A+B) vs Delta(A)+Delta(B) cannot be tested. The uricase experiment is a depletion, not a mixture.

Phase 11 — controls

```
unspiked serum: replicate cos 0.9988, OOD 0.282
ILS blanks (n=225): replicate cos 0.594, max batch drift 0.0236 — no systematic drift
uricase depletion: |Δ| 0.250, cos vs urate direction -0.608 (negative = chemically expected)
urate vs isotopic urate: distance 0.196, cos 0.868
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Serum spike test, limitations and next steps

Phase 7 — serum spike versus pure analyte

matched cos (median)	-0.0124
mismatched null (median)	-0.0361
median angle	90.7 deg
median distance ratio	0.549
analytes beating 95th pct null	7/53

Analytes with directional agreement

analyte	family	conc µM	cos	angle	|Δ|
hypoxanthine	purine	10.0	0.889	27.3	0.114
xanthine	purine	50.0	0.736	42.6	0.120
creatinine	small_nitrogenous	80.0	0.584	54.2	0.018
ergothioneine	cofactor	5.0	0.534	57.7	0.098
ascorbate	organic_acid	60.0	0.481	61.2	0.015
guanine	purine	10.0	0.402	66.3	0.080
acetoacetate	organic_acid	10.0	0.348	69.6	0.047
glycerol	polyol	40.0	0.256	75.2	0.021
proline	amino_acid	168.0	0.172	80.1	0.040
leucine	amino_acid	125.0	0.132	82.4	0.015

Limitations

• Every perturbation dataset is Ag/Au-SERS projected into a Raman atlas; nothing here validates in-domain Raman behaviour.

• Each serum spike has a single concentration, so no in-serum dose-response and no combinatorial mixture test (Phase 9) is possible.

• Distance-from-control can increase for reasons unrelated to the analyte (colloid aggregation state, laser power, surface fouling). The permutation null controls for concentration-label assignment but not for a confound that varies monotonically with concentration by design.

• Prior GAIRA work established these Ag-SERS spectra are background-dominated; the serum-spike null is consistent with that and should not be read as a property of the Raman atlas.

Recommended next experiments

1. An in-domain Raman dose-response (pure analyte, Raman, several concentrations) to separate 'the atlas cannot track concentration' from 'Ag-SERS is the limiting factor'. No such dataset currently exists in GAIRA and this is the single most informative missing measurement.

2. Serum spikes at a concentration series rather than one level, so in-serum dose-response and detection thresholds can be estimated per analyte.

3. Blank-colloid difference acquisition, to test directly whether background suppression restores directional agreement for weak adsorbers.